CONTENT-BASED RECOMMENDATION SYSTEM

Auspify Internship Task Report

1. Introduction

This project was completed as part of the www.Auspify.com internship task. The objective was to develop a content-based recommendation system capable of recommending titles that are similar to a selected title based on their content characteristics.

The project followed a five-step workflow:

Extract relevant content features.

Perform text preprocessing.

Calculate content similarity.

Generate recommendations for selected titles.

Evaluate recommendation quality.

The system was developed using Python and data science libraries, with the goal of demonstrating the complete process of building and evaluating a recommendation system.

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2. Project Objective

The main objective of this project was to build a recommendation system that can identify titles with similar content and provide relevant recommendations to users.

The project also aimed to:

Prepare and clean the dataset.

Extract useful content features.

Process textual data for analysis.

Calculate similarity between titles.

Generate top-N recommendations.

Evaluate the quality of the recommendations.

Export the cleaned data-set for future use.

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3. Methodology

Step 1: Extract Relevant Content Features

The first stage involved examining the dataset and identifying features that could be used to determine similarities between titles.

Relevant content features included title-related information and categorical/textual information such as the `listed_in` field.

These features were selected because they provide information about the type and category of each title and can be used to identify titles with similar characteristics.

The data set was also inspected for data-quality issues before being used by the recommendation system.

Step 2: Perform Text Pre-processing

The textual data was pre-processed before calculating similarity.

The pre-processing process included operations such as:

Converting text to lowercase.

Handling missing or empty values.

Cleaning unnecessary spaces.

Standardizing textual values.

Preparing the selected content features for similarity analysis.

Text pre-processing was important because inconsistent text formatting can negatively affect similarity calculations.

For example, converting text to lowercase ensures that values such as:

`Drama`

and

`drama`

are treated consistently.

Step 3: Calculate Content Similarity

After pre-processing the data-set, the relevant content features were converted into a numerical representation that could be compared mathematically.

A similarity matrix was generated to measure how closely titles are related to one another.

The recommendation system uses similarity scores to identify the titles that are most similar to a selected title.

A higher similarity score indicates a stronger relationship between two titles.

For example, recommendations generated for Gang lands included:

Recommended Title	Similarity
Sentinelle	0.759
Lupin	0.730
Crime Time	0.730
Earth and Blood	0.688
Taken	0.677

The original title itself was excluded from its recommendation list.

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4. Recommendation Generation

Step 4: Generate Recommendations for Selected Titles

A recommendation function was developed to accept a title and return the top five most similar titles.

The system was tested using several selected titles:

Ganglands

Midnight Mass

Sankofa

Dick Johnson Is Dead

The recommendation function searches for the selected title, retrieves its similarity scores, sorts the other titles according to their similarity, and returns the top five recommendations.

This allows the system to automatically generate recommendations based on the content characteristics of the selected title.

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5. Recommendation Quality Evaluation

Step 5: Evaluate Recommendation Quality

The recommendation system was evaluated using **Precision@5**.

Precision@5 measures how many of the five recommended titles were considered relevant according to the relevance criterion used in the project.

The evaluation produced the following results:

Test Title	Precision @ 5
Ganglands	60.0%
Midnight Mass	100.0%
Sankofa	100.0%
Dick Johnson Is Dead	0.0%
Average	**65.0%**

Interpretation

The system achieved an overall average Precision@5 of **65.0%** across the selected test titles.

The highest scores were achieved by **Midnight Mass** and **Sankofa**, both recording 100% Precision@5.

Ganglands achieved 60%, indicating that a portion of its recommendations matched the relevance criterion.

Dick Johnson Is Dead recorded 0%. This does not necessarily mean that all recommendations were objectively unrelated. Rather, it indicates that the recommendations did not satisfy the specific relevance criterion used for this evaluation.

The results demonstrate that the recommendation system can successfully identify relevant content for several titles, while also highlighting areas where the recommendation model could be improved.

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6. Data Cleaning and Export

Before using the dataset for the recommendation system, the data was processed and cleaned to improve consistency and reliability.

The cleaned dataset was exported after preprocessing so that it can be reused for future analysis and development.

The exported file is:

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cleaned_dataset.csv
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The cleaned dataset serves as the processed version of the original data used throughout the project.

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7. Technologies Used

The following technologies and libraries were used during the project:

Python — Primary programming language.

Pandas — Data loading, cleaning, transformation, and analysis.

NumPy — Numerical operations.

Scikit-learn — Text/vector processing and similarity calculations.

Jupyter Notebook / VS Code — Development and experimentation.

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8. Challenges Encountered

Several challenges were encountered during development, including:

Preparing textual data for similarity analysis.

Handling inconsistent text values.

Ensuring that the selected title could be correctly located in the dataset.

Implementing the recommendation function correctly.

Evaluating recommendations using a measurable metric.

Debugging errors during implementation.

These challenges provided practical experience in data preprocessing, Python programming, debugging, machine learning concepts, and recommendation-system development.

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9. Results

The project successfully produced a working content-based recommendation system.

The system is capable of:

Processing the dataset.

Extracting relevant content features.

Pre-processing textual information.

Calculating content similarity.

Generating the top five recommendations for a selected title.

Measuring recommendation quality using Precision@5.

Exporting the cleaned dataset.

The evaluation produced an average Precision@5 of:

65.0%

This demonstrates that the system was able to produce relevant recommendations for a significant portion of the tested titles.

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10. Conclusion

The Auspify internship task was successfully completed by implementing a content-based recommendation system through the required five-step workflow.

Relevant content features were extracted from the dataset, the textual data was preprocessed, similarity between titles was calculated, recommendations were generated for selected titles, and the recommendation quality was evaluated using Precision@5.

The system achieved an average Precision@5 of **65.0%** across the selected test titles.

The cleaned dataset was also exported for future analysis and reuse.

Overall, the project provided practical experience in data preprocessing, feature engineering, text processing, similarity-based recommendation systems, model evaluation, Python programming, and data science workflow implementation.